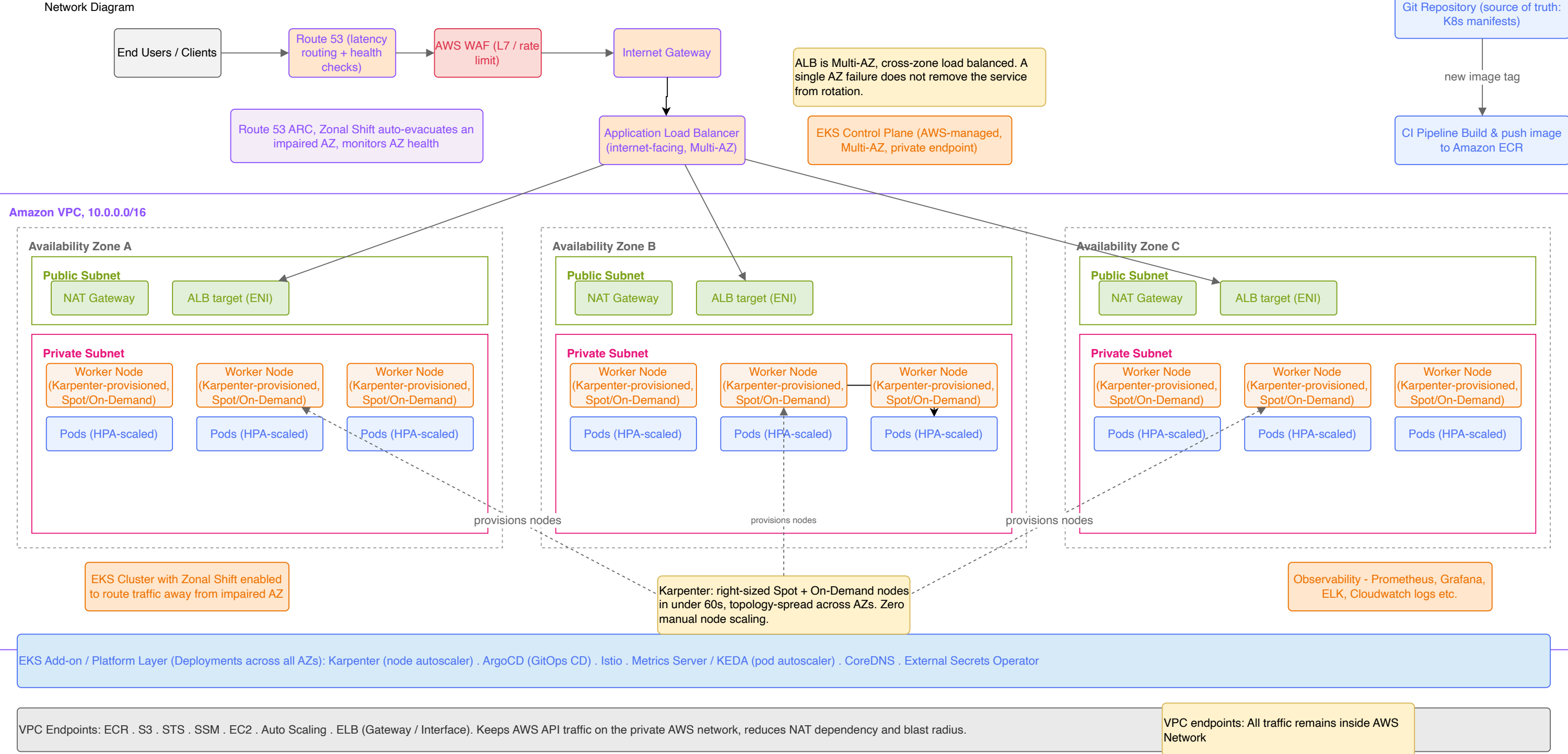


AWS EKS Architecture



CloudWatch (metrics, logs, alarms)

X-Ray (distributed tracing)

Systems Manager (patching, session mgr)

Auto Scaling

KMS (encryption at rest)

IAM / IRSA (least-privilege roles)

Secrets Manager

Security, Least Privilege & Defense in Depth

- EKS API server: private endpoint, no public access
- Security Groups: least privilege, scoped to VPC CIDR, TLS only
- IRSA / EKS Pod Identity: per-pod least-privilege IAM
- KMS: encrypts EKS secrets (etcd), EBS, DynamoDB, ElastiCache
- AWS WAF at the edge: OWASP Top 10 rules + rate limiting
- Secrets Manager + External Secrets Operator for app secrets

Reliability, Spikes, Zero Manual Intervention

- HPA / KEDA scale pods on CPU + queue depth within seconds
- Karpenter scales nodes to match pending pods, Spot+On-Demand mix
- PodDisruptionBudgets + topologySpreadConstraints keep quorum through node churn, AZ loss, or rolling deploys
- ArgoCD auto-sync + self-heal reverts bad or drifted deployments
- Route 53 ARC zonal shift evacuates an impaired AZ in minutes